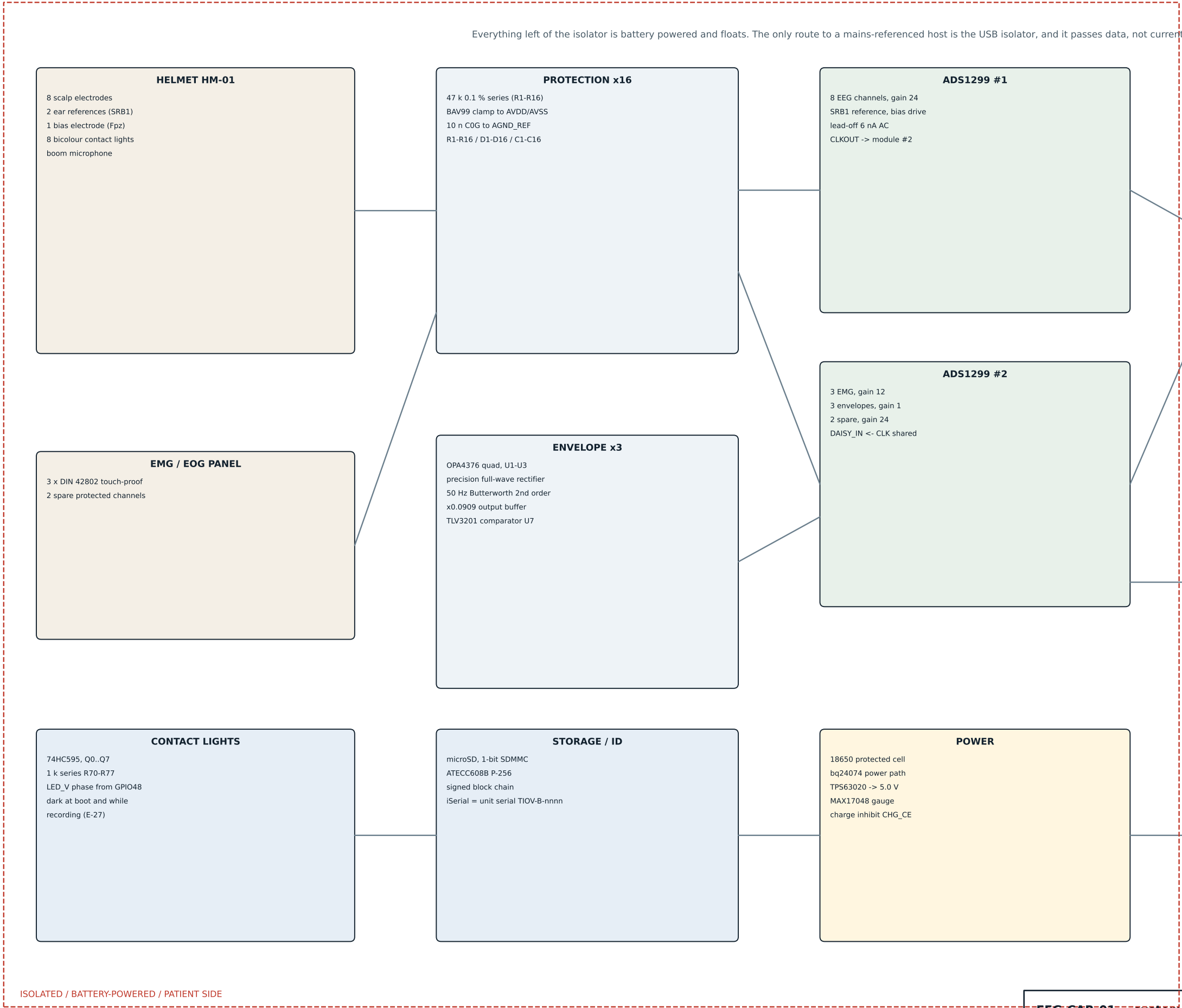
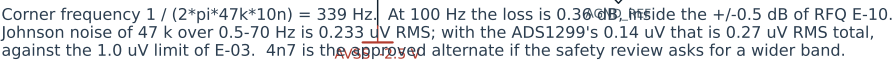


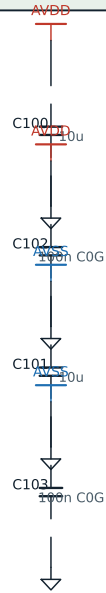
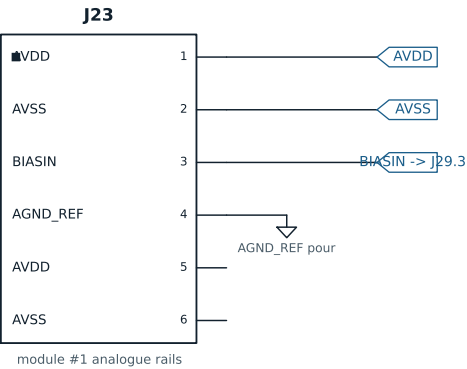
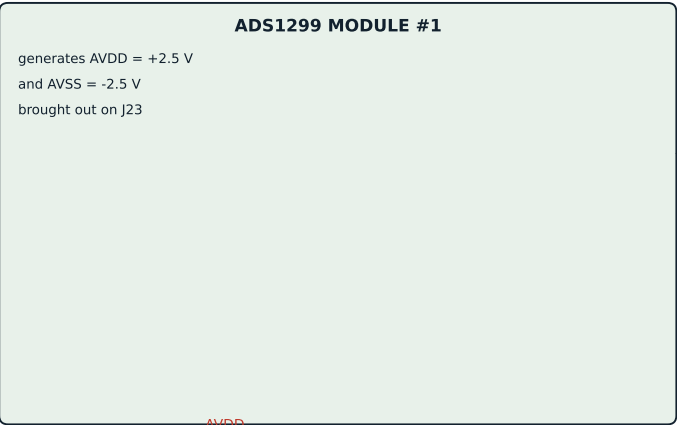


ISOLATED / BATTERY-POWERED / PATIENT SIDE

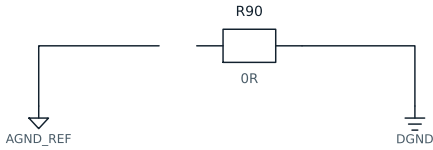


R11 runs the other way: BIASOUT leaves ADS1299 module #1 at J2.10, passes through R11 and reaches the Fpz bias electrode at J14.11. It is a driven common-mode return, not an earth. D11 and C11 sit on the electrode side.

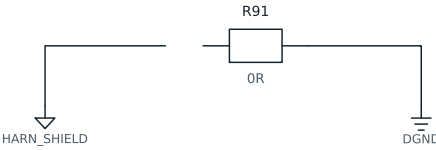
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THE SINGLE STAR POINT

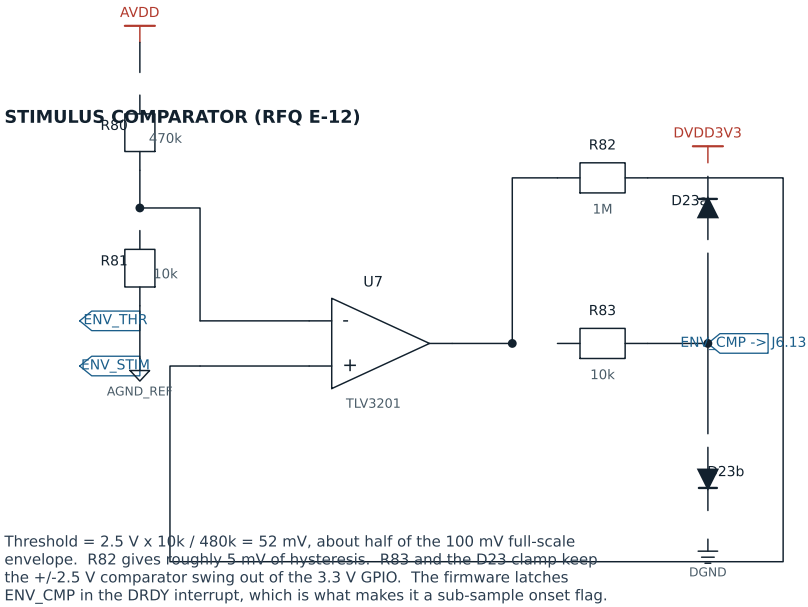


R90 is the ONLY connection between the analogue reference and digital ground. It is a 0 ohm 0603 part, not a copper bridge, so it can be lifted for a leakage measurement and refitted.



R91 grounds the helmet cable screen at the pod end only, so the screen carries no loop current. The far end of the screen is left open at the electrode assemblies (WH-EEG-008 section 3).

AGND_REF IS NOT GROUND. With AVDD = +2.5 V and AVSS = -2.5 V, AGND_REF is the analogue mid-rail and the reference every electrode, clamp and op-amp input is measured against. It is poured on both layers over the whole analogue zone and is the guard either side of every input trace.



EEG-CAR-01 -- analogue rails, star grounds and the stimulus comparator

Where the analogue reference is made, and the one place it meets digital ground

J1	ADS1299 module #1 digital
1	DVDD3V3
2	DGND
3	SCLK
4	MOSI
5	MISO
6	DRDY
7	CS
8	START
9	RESET
10	CLK_ADS
11	V5V
12	DGND

J2	ADS1299 module #1 analogue signals
1	IN1
2	IN2
3	IN3
4	IN4
5	IN5
6	IN6
7	IN7
8	IN8
9	SRB1
10	BIASOUT

J23	ADS1299 module #1 analogue rails
1	AVDD
2	AVSS
3	BIASIN
4	AGND_REF
5	AVDD
6	AVSS

J3	ADS1299 module #2 digital
1	DVDD3V3
2	DGND
3	SCLK
4	MOSI
5	DAISY
6	NC_DRDY2
8	START
9	RESET
10	CLK_ADS
11	V5V
12	DGND

J4	ADS1299 module #2 analogue signals
1	EMG1
2	EMG2
3	EMG3
4	ENV_STIM
5	ENV_VOICE
6	ENV_ROOM
7	SPARE1
8	SPARE2
9	SRB1
10	NC_BIASOUT2

J29	ADS1299 module #2 analogue rails
1	AVDD2
2	AVSS2
3	BIASIN
4	AGND_REF
5	AVDD2
6	AVSS2

J5	DAISY / clock stub
1	DAISY
2	CLK_ADS
3	DGND
4	DGND

J14	helmet harness, electrodes
1	E_Fz
2	E_Cz
3	E_Pz
4	E_C3
5	E_C4
6	E_T7
7	E_T8
8	E_F7
9	REF_L
10	REF_R
11	BIAS_EL
12	HARN_SHIELD

J30	helmet harness, contact lights
1	LED1
2	LED2
3	LED3
4	LED4
5	LED5
6	LED6
7	LED7
8	LED8
9	LED_V
10	LED_GND

J22	EOG / spare header
1	EOGIN1
2	AGND_REF
3	EOGIN2

Module connectors are NOT direct plug-in sockets. The purchased modules sit on the printed module plate MP-01 above the carrier and are joined with keyed 2.54 mm ribbon jumpers per ICD-EEG-006. The one exception is the ESP32-S3-DevKitC-1, which is inserted directly into J6 and J7 (row spacing 22.86 mm).

J6	ESP32-S3-DevKitC-1 row A
1	DVDD3V3
2	DVDD3V3
3	RESET_EN
4	BTN_A
5	BTN_B
6	BTN_STOP
7	I2S_DIN
8	START
9	RESET
10	I2S_MCLK
11	I2S_BCLK
12	I2S_LRCK
13	ENV_CMP
14	VBUS_DET
15	I2S_DOUT
16	CS
17	MOSI
18	SCLK
19	MISO
20	DRDY
21	V5V
22	DGND

J7	ESP32-S3-DevKitC-1 row B
1	DGND
2	UART_TX
3	UART_RX
4	SDA
5	SCL
6	LED_SR_CLK
7	LED_SR_DATA
8	SD_D0
9	SD_CLK
10	SD_CMD
11	NC_GPIO37
12	NC_GPIO36
13	NC_GPIO35
14	LED_SR_LATCH
15	NC_GPIO45
16	LED_PWM
17	CHG_CE
18	MIC_MUTE
19	NC_GPIO12 and 13 are GPIO37, 36 and 35. They carry the octal PSRAM on the -N16R8 variant and MUST be left unconnected; position 15 is GPIO45, the VDD_SPI strapping pin, also open. The microSD interface is one-bit SDMMC. 70 kB/s is needed and about 2 MB/s is available; the three data lines that four-bit mode would have used are spent on the contact-light shift register instead.
20	USB_DN
21	DGND
22	DGND

J8	ES8388 codec module
1	DVDD3V3
2	DGND
3	I2S_MCLK
4	I2S_BCLK
5	I2S_LRCK
6	I2S_DIN
7	I2S_DOUT
8	SDA
9	SCL
10	HP_TAP
11	HP_L
12	HP_R
13	HP_GND
14	V5V

J9	codec microphone feeds
1	VOICE_PRE
2	ROOM_PRE
3	DGND
4	MIC_MUTE

J10	ADuM4160 isolator, device side
1	VDD_ISO
2	USB_DN
3	USB_DP
4	DGND

J11	ATECC608B breakout
1	DVDD3V3
2	DGND
3	SDA
4	SCL

J12	charger + gauge module
1	VBAT
2	DGND
3	VBUS_CHG
4	CHG_CE
5	SDA
6	SCL
7	VSYS
8	NC_CHG_STAT

J20	microSD breakout
1	DVDD3V3
2	DGND
3	SD_CLK
4	SD_CMD
5	SD_D0
6	DGND
7	DGND
8	DGND

J21	boom preamp (part open, E-14)
1	DVDD3V3
2	DGND
3	VOICE_PRE
4	VOICE_RAW
5	NC_MIC_GAIN
6	DGND

J18	boom microphone pigtail
1	VOICE_RAW
2	DGND
3	DGND
4	DGND

J28	room microphone module
1	DVDD3V3
2	DGND
3	ROOM_PRE
4	MIC_MUTE

J27	headphone jack pigtail
1	HP_L
2	HP_R
3	HP_GND
4	NC_HP_DET

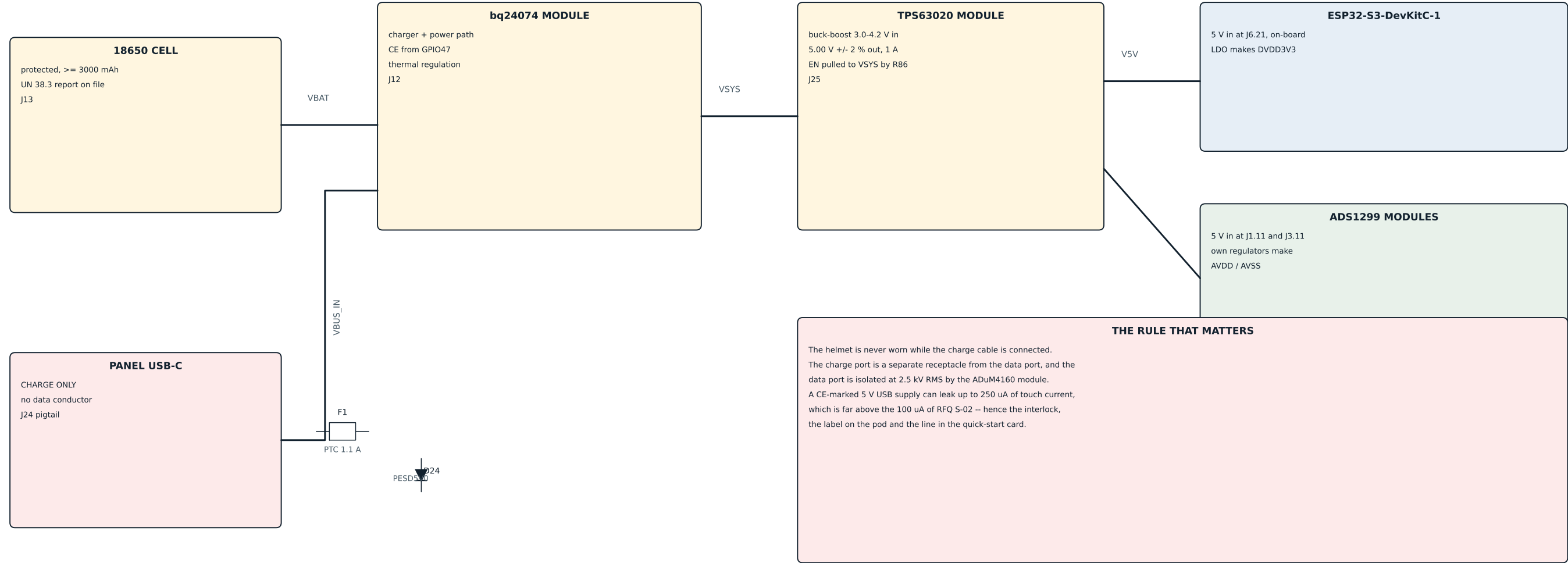
J25	buck-boost module
1	VSYS
2	DGND
3	DGND
4	DGND
5	BOOST_EN
6	NC_BOOST_PG

J26	debug / programming
1	DVDD3V3
2	DGND
3	UART_TX
4	UART_RX
5	RESET_EN
6	NC_GPIO0

J24	charge input pigtail
1	VBUS_IN
2	DGND

J13	battery
1	VBAT
2	DGND

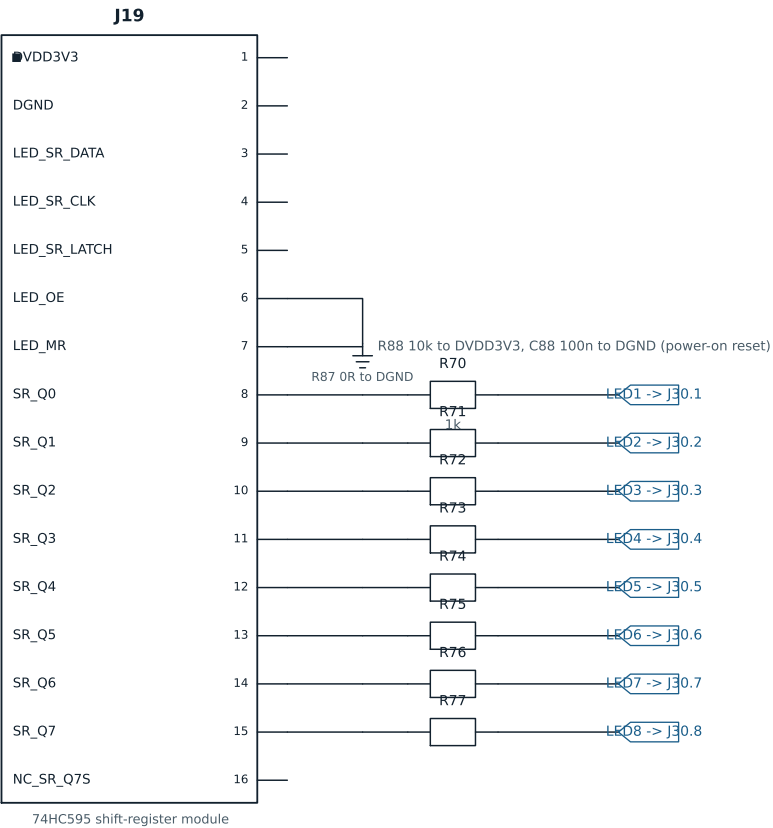
EEG-CAR-01 -- digital and power connectors	
Pin-by-pin assignment, generated from the netlist	
TI One Voice research programme -- CC BY-SA 4.0	Rev B 2026-09-02 Sheet 6 of 8



VBUS_DET divider R84 100k / R85 56k -> 1.79 V at GPIO46. The firmware refuses CMD_START_SESSION while VBUS_DET is high, and the charger enable CHG_CE is driven low for the whole of a session, so charging cannot start while recording either. Two independent mechanisms, one requirement: RFQ S-01.

RAIL	VALUE	SOURCE
VBAT	3.0 - 4.2 V	cell terminal, J13.1 -> J12.1
VSYS	3.2 - 4.4 V	charger power-path output, J12.7
V5V	5.00 V, 1 A	buck-boost output; DevKit and both ADS modules
DVDD3V3	3.30 V, 0.5 A	DevKit on-board LDO; all 3.3 V logic
VDD_ISO	3.30 V	isolator device side, through ferrite L1
AVDD	+2.50 V	from ADS1299 module #1, analogue only
AVSS	-2.50 V	from ADS1299 module #1, analogue only
AGND_REF	0 V	analogue mid-rail; joins DGND at R90 only
DGND	0 V	digital return

ECO-EEG-002: Rev A had no source for DVDD3V3 at all -- the charger's SYS output, the DevKit's 5 V pin and the ADS module supplies were all unconnected nets. The board could not have powered up. Rev B adds J25, F1, D24, R84-R86 and C70-C74.



ECO-EEG-001: in Rev A the eight LEDn nets existed on the harness connector with nothing driving them -- the 74HC595 module's outputs were not brought to the carrier at all, so the contact lights could not work. Rev B widens J19 to 1x16, adds R70-R77 and defines the phase scheme above. ECO-EEG-014 moves the light lines out of the electrode harness into their own cable (J30).

HOW THREE COLOURS COME OUT OF EIGHT LINES

Each site carries one two-lead bicolour LED: red and green dice in inverse parallel between its LEDn line and the LED_V common.

Phase A: LED_V driven high, Qn low -> that site shows green.

Phase B: LED_V driven low, Qn high -> that site shows red.

Alternating the two phases above 100 Hz shows amber.

A site that is off in both phases is dark. All eight are independent, and no extra conductor is needed.

DARK AT BOOT, DARK WHILE RECORDING

LED_V is GPIO48. At reset it is an input, so it floats and no current can flow through any LED whatever the shift register happens to contain. That is the power-on guarantee.

During a recording block the firmware holds LED_V low and clears the register (RFQ E-27), so nothing on the head can be watched or reacted to while data is being taken.

WHAT THE LIGHTS MEAN

green below 10 kohm ready, do nothing

amber 10 - 20 kohm add gel through that site's port

red above 20 kohm slide the assembly a few mm, then gel

pulse measurement running, keep still

off recording, or the instrument is not powered

Driven from the ADS1299's own lead-off measurement, so the head and the screen can never disagree.

EEG-CAR-01 -- contact lights

Eight two-lead bicolour LEDs on eight lines and one phase common